Analytical Chemistry Cumulative Examination
16th April 2004

Professor Susan Ziegler of the University of Arkansas' Biological Sciences Department recently presented a seminar in which she discussed the role of stable isotope mass spectrometry in environmental sciences and the role of biological organisms in determining $^{13}C/^{12}C$ ratios of soil and water.

Using the attached excerpt from the Chart of the Nuclides, discuss the following:

The synthesis of ^{12}C.

The synthesis of ^{13}C.

The synthesis of ^{14}C.

Discuss factors leading to the variations in the ratios of $^{13}C/^{12}C$ and $^{14}C/^{12}C$ and how these variations are used in the scientific exploration of the world and cosmos around us. Useful examples can be obtained from Professor Zielger's talk and from meteorite studies. Credit will be given for quantitative answers that make good use of examples, the literature, diagrams, the Chart of the Nuclides.